

DESIGN CODE OF HALF ADDER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/ha.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 07/30/2026 05:10:46 PM
6	-- Design Name:
7	-- Module Name: ha - Behavioral
8	-- Project Name:
9	-- Target Devices:
10	-- Tool Versions:
11	-- Description:
12	--
13	-- Dependencies:
14	--
15	-- Revision:
16	-- Revision 0.01 - File Created
17	-- Additional Comments:
18	--
19	-----
	--
20	
21	
22	library IEEE;
23	use IEEE.STD_LOGIC_1164.ALL;
24	
25	-- Uncomment the following library declaration if using
26	-- arithmetic functions with Signed or Unsigned values
27	--use IEEE.NUMERIC_STD.ALL;
28	
29	-- Uncomment the following library declaration if instantiating
30	-- any Xilinx leaf cells in this code.
31	--library UNISIM;
32	--use UNISIM.VComponents.all;
33	
34	entity ha is
35	Port (A : in STD_LOGIC;
36	B : in STD_LOGIC;
37	SUM : out STD_LOGIC;
38	Cout : out STD_LOGIC);
39	end ha;
40	
41	architecture dataflow of ha is
42	
43	begin
44	sum <= a xor b;
45	cout <= a and b;
46	

47	
48	end dataflow;

DESIGN CODE OF FULL ADDER

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/fa.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 08/06/2026 04:18:02 PM
6	-- Design Name:
7	-- Module Name: fa - Behavioral
8	-- Project Name:
9	-- Target Devices:
10	-- Tool Versions:
11	-- Description:
12	--
13	-- Dependencies:
14	--
15	-- Revision:
16	-- Revision 0.01 - File Created
17	-- Additional Comments:
18	--
19	-----
	--
20	
21	
22	library IEEE;
23	use IEEE.STD_LOGIC_1164.ALL;
24	
25	-- Uncomment the following library declaration if using
26	-- arithmetic functions with Signed or Unsigned values
27	--use IEEE.NUMERIC_STD.ALL;
28	
29	-- Uncomment the following library declaration if instantiating
30	-- any Xilinx leaf cells in this code.
31	--library UNISIM;
32	--use UNISIM.VComponents.all;
33	
34	entity fa is
35	Port (a : in STD_LOGIC;
36	b : in STD_LOGIC;
37	cin : in STD_LOGIC;
38	sum : out STD_LOGIC;
39	cout : out STD_LOGIC);
40	end fa;
41	
42	architecture dataflow of fa is
43	

44	begin
45	sum <= a xor b xor cin;
46	cout <= (a and b) or (b and cin) or (cin and a);
47	
48	end dataflow;

DESIGN CODE OF RCA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/rca8.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 08/06/2026 04:21:49 PM
6	-- Design Name:
7	-- Module Name: rca8 - Behavioral
8	-- Project Name:
9	-- Target Devices:
10	-- Tool Versions:
11	-- Description:
12	--
13	-- Dependencies:
14	--
15	-- Revision:
16	-- Revision 0.01 - File Created
17	-- Additional Comments:
18	--
19	-----
	--
20	
21	
22	library IEEE;
23	use IEEE.STD_LOGIC_1164.ALL;
24	
25	-- Uncomment the following library declaration if using
26	-- arithmetic functions with Signed or Unsigned values
27	--use IEEE.NUMERIC_STD.ALL;
28	
29	-- Uncomment the following library declaration if instantiating
30	-- any Xilinx leaf cells in this code.
31	--library UNISIM;
32	--use UNISIM.VComponents.all;
33	
34	entity rca8 is
35	Port (a : in STD_LOGIC_VECTOR (7 downto 0);
36	b : in STD_LOGIC_VECTOR (7 downto 0);
37	sum : out STD_LOGIC_VECTOR (7 downto 0);
38	cout : out STD_LOGIC);
39	end rca8;
40	

41	architecture structural of rca8 is
42	
43	component ha
44	Port (A : in STD_LOGIC;
45	B : in STD_LOGIC;
46	SUM : out STD_LOGIC;
47	Cout : out STD_LOGIC
48	);
49	end component;
50	
51	component fa
52	Port (a : in STD_LOGIC;
53	b : in STD_LOGIC;
54	cin : in STD_LOGIC;
55	sum : out STD_LOGIC;
56	cout : out STD_LOGIC);
57	end component;
58	
59	signal cint : std_logic_vector(6 downto 0);
60	
61	begin
62	
63	inst1 : ha port map(a => a(0) ,b => b(0) ,sum => sum(0) ,cou
64	t => cint(0));
64	inst2 : fa port map(a => a(1) ,b => b(1) ,cin => cint(0) ,sum => sum(1) ,cou
65	t => cint(1));
65	inst3 : fa port map(a => a(2) ,b => b(2) ,cin => cint(1) ,sum => sum(2) ,cou
66	t => cint(2));
66	inst4 : fa port map(a => a(3) ,b => b(3) ,cin => cint(2) ,sum => sum(3) ,cou
67	t => cint(3));
67	inst5 : fa port map(a => a(4) ,b => b(4) ,cin => cint(3) ,sum => sum(4) ,cou
68	t => cint(4));
68	inst6 : fa port map(a => a(5) ,b => b(5) ,cin => cint(4) ,sum => sum(5) ,cou
69	t => cint(5));
69	inst7 : fa port map(a => a(6) ,b => b(6) ,cin => cint(5) ,sum => sum(6) ,cou
70	t => cint(6));
70	inst8 : fa port map(a => a(7) ,b => b(7) ,cin => cint(6) ,sum => sum(7) ,cou
71	t => cout);
71	
72	
73	end structural;

TEST BENCH CODE OF RCA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/rca8_tb.vhd [VHDL]	
1	-----
	--
2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 08/13/2026 04:13:11 PM
6	-- Design Name:
7	-- Module Name: rca8_tb - testing

```

8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20 --
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity rca8_tb is
35 -- Port ( );
36 end rca8_tb;
37
38 architecture testing of rca8_tb is
39
40     component rca8 is
41         Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
42               b : in STD_LOGIC_VECTOR (7 downto 0);
43               sum : out STD_LOGIC_VECTOR (7 downto 0);
44               cout : out STD_LOGIC);
45     end component;
46
47
48     signal a : std_logic_vector(7 downto 0);
49     signal b : std_logic_vector(7 downto 0);
50     signal s : std_logic_vector(7 downto 0);
51     signal c : std_logic;
52
53
54
55 begin
56
57
58     rca_test : rca8 port map(

```

59	a => a,
60	b => b,
61	sum => s,
62	cout =>c
63	);
64	
65	
66	process
67	
68	begin
69	a <= "01010101";
70	b <= "10101010";
71	
72	wait for 10ns;
73	
74	a <= "00110011";
75	b <= "11001100";
76	
77	wait for 10ns;
78	
79	a <= "10101010";
80	b <= "11001100";
81	
82	wait for 10ns;
83	
84	a <= "00110001";
85	b <= "10011100";
86	
87	wait for 10ns;
88	
89	a <= "11001010";
90	b <= "11000011";
91	
92	wait for 10ns;
93	
94	a <= "11111111";
95	b <= "11111111";
96	
97	wait for 10ns;
98	
99	a <= "00000000";
100	b <= "00000000";
101	
102	wait for 10ns;
103	
104	a <= "01011010";
105	b <= "11110011";
106	
107	wait for 10ns;
108	
109	a <= "00000001";

110	b <= "10000111";
111	
112	wait for 10ns;
113	
114	a <= "10101010";
115	b <= "11111111";
116	
117	
118	wait;
119	
120	end process;
121	
122	
123	
124	end testing;

SCREENSHOT OF CODE

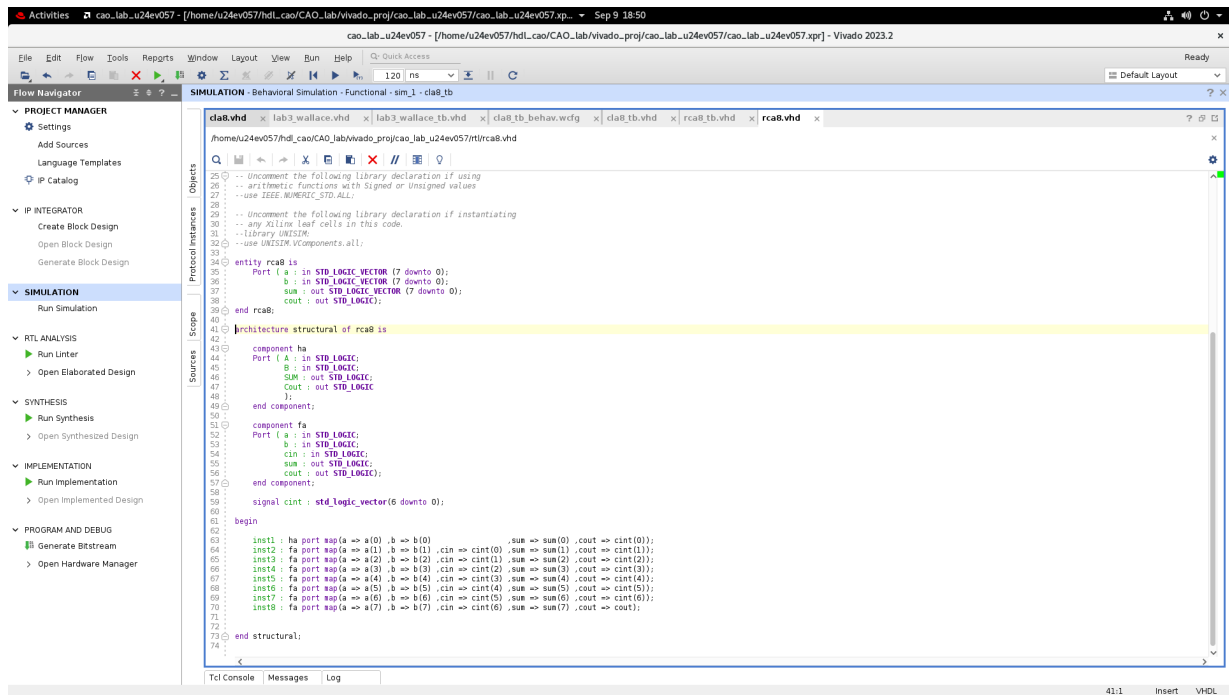


Figure 1: Design

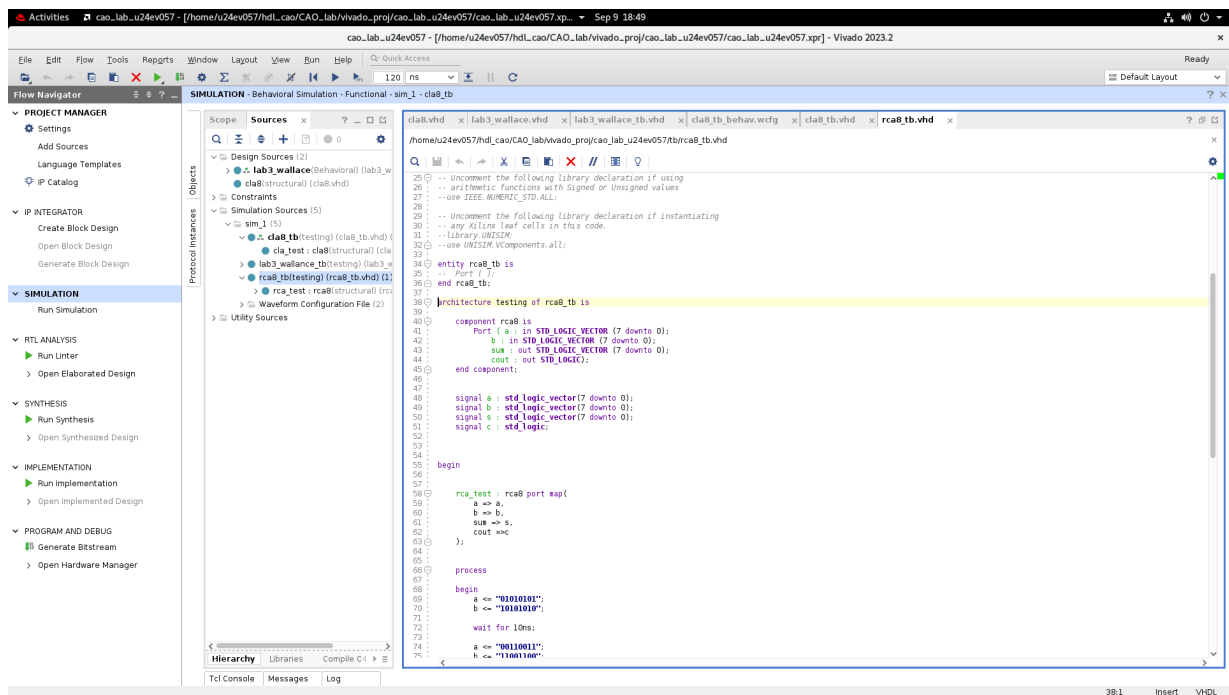


Figure 2: Tb

WAVEFORM

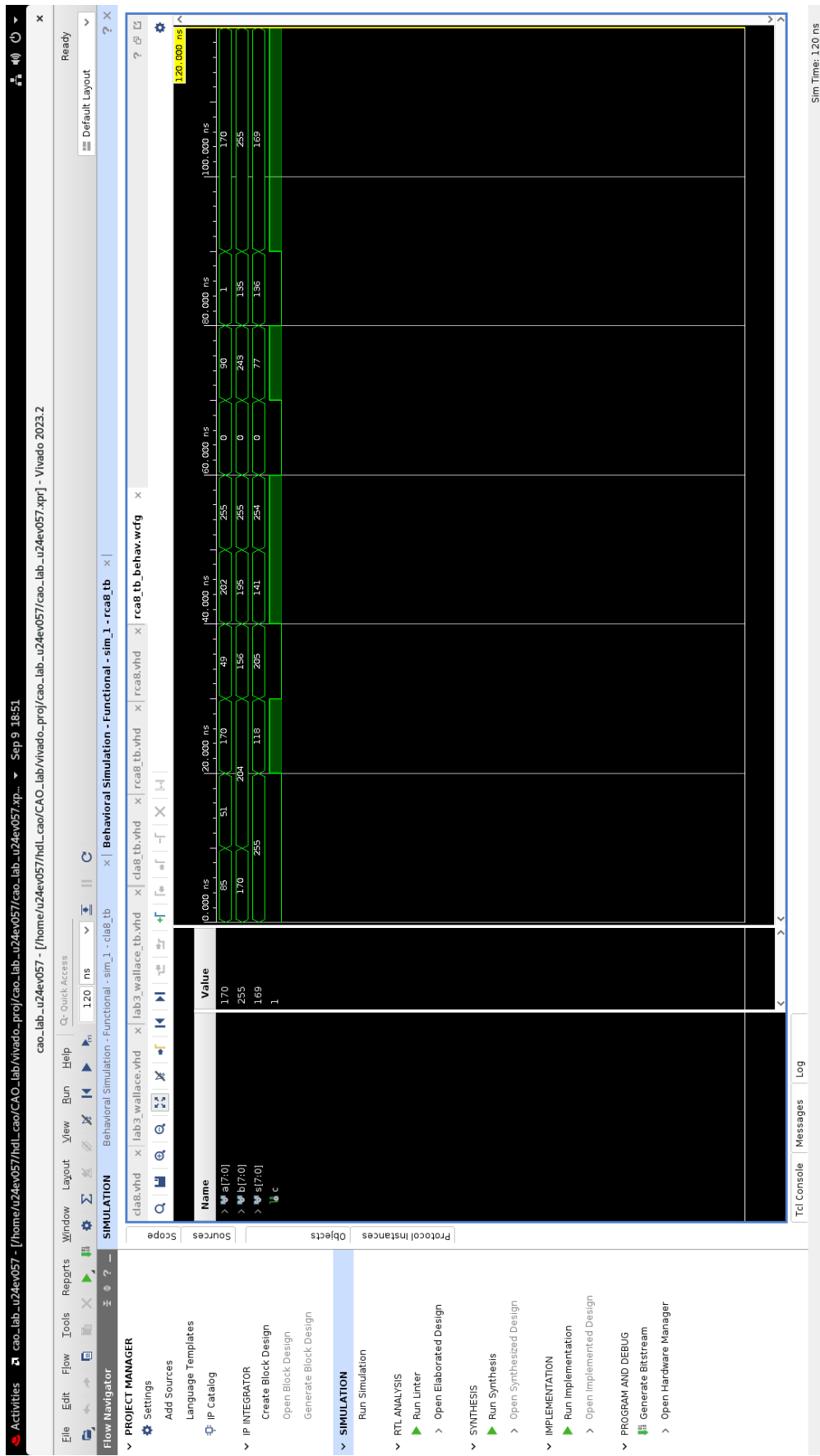


Figure 3: Waveform

